Huawei HEB-ACC-42 Access Node QoS Congestion Event Resolution SOP

Node: HEB-ACC-42 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5019

Description

The QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) failure domain in the Huawei Access Node HEB-ACC-42 at Hebbal Edge is primarily attributed to the inherent complexity of the Quality of Service (QoS) mechanism, which is designed to manage and prioritize traffic flow across the network. The Huawei Access Node, as a critical component of the network infrastructure, relies on the QoS mechanism to ensure efficient and effective transmission of traffic. However, this very mechanism can sometimes lead to congestion events, such as microbursts and interface tail drops, due to the dynamic nature of traffic flow and the limited buffer size of the QoS queues. Furthermore, the Huawei Access Node's Role as an Access device, which is responsible for aggregating and forwarding traffic from multiple sources, can exacerbate the likelihood of QoS queue starvation, resulting in a Critical alarm code ALARM_5019 being triggered.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and all necessary physical connections are secure.
- Check the redundant link status to ensure that it is stable and functional.
- Verify that the Huawei Access Node HEB-ACC-42 is powered on and all fans are operational.

Potential Root Causes

Cause	Probability
QoS queue starvation due to excessive traffic load	High
Microbursts caused by sudden spikes in traffic flow	Medium
Interface tail drops due to buffer overflow	Low

Diagnostic Commands & Expected Outputs

```
Command: display queue-status interface GigabitEthernet 0/0/0, Expected Output: Queue statistics for GigabitEthernet 0/0/0: Queue 1: drops 100, bytes 5000 Queue 2: drops 0, bytes 0, Description: This command displays the queue statistics for the specified interface, allowing the operator to identify any QoS queue starvation or drops.

Command: display traffic-statistics interface GigabitEthernet 0/0/0, Expected Output: Traffic statistics for GigabitEthernet 0/0/0: Tx packets 1000, bytes 500000 Rx packets 500, bytes 20000, Description: This command displays the traffic statistics for the specified interface, enabling the operator to detect any microbursts or sudden spikes in traffic flow.
```

Resolution Steps

Step 1: Step: Adjust QoS queue configuration, Command: queue-adjust interface GigabitEthernet 0/0/0 queue 1 threshold 50, Description: This command adjusts the QoS queue configuration for the specified interface, allowing the operator to increase the threshold for drops and prevent QoS queue starvation.

Step 2: Step: Clear interface statistics, Command: clear interface statistics GigabitEthernet 0/0/0, Description: This command clears the interface statistics for the specified interface, enabling the operator to reset the traffic flow and prevent any further congestion events.

Post-Check Verification

```
Command: display queue-status interface GigabitEthernet 0/0/0, Expected Output: Queue statistics for GigabitEthernet 0/0/0: Queue 1: drops 0, bytes 0, Description: This command verifies that the QoS queue starvation has been resolved and no further drops are occurring.

Command: display traffic-statistics interface GigabitEthernet 0/0/0, Expected Output: Traffic statistics for GigabitEthernet 0/0/0: Tx packets 0, bytes 0 Rx packets 0, bytes 0, Description: This command verifies that the microbursts and sudden spikes in traffic flow have been resolved and no further traffic is being dropped.
```

Roll-back Procedures

- Step: Restore original QoS queue configuration, Command: queue-restore interface GigabitEthernet 0/0/0 queue 1 threshold 10, Description: This command restores the original QoS queue configuration for the specified interface, reverting any changes made during the resolution steps.

- Step: Clear interface statistics, Command: clear interface statistics GigabitEthernet 0/0/0, Description: This command clears the interface statistics for the specified interface, enabling the operator to reset the traffic flow and prevent any further congestion events.